# Notice, question, predict



**antocciphysics | TEACHER KEY | CPA + MATCHING SEI**

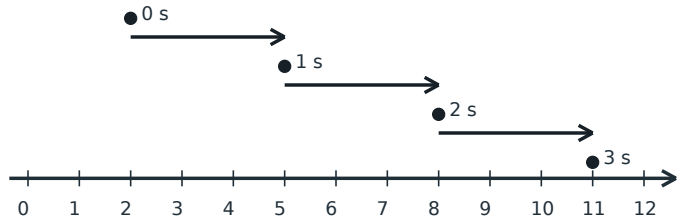
**Learning target:** Use observations to make a prediction that can be tested.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

**01. Read a scale**
Marks 5 cm apart; rightward arrow from 2 to 7 cm.

**02. Worked example: explain the model**
Four equally spaced points; the 11 m point is the prediction. A new measurement could disagree.

**Model / diagram check**

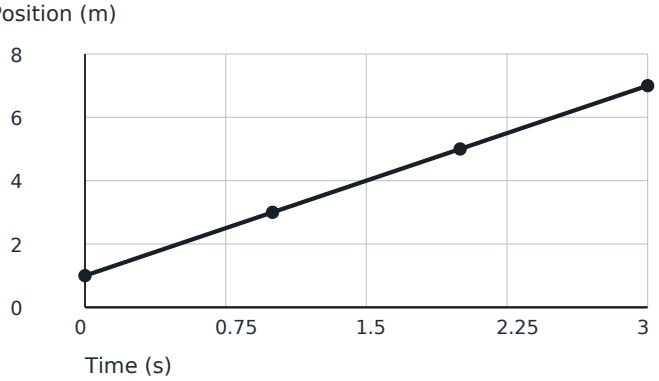



Position (m); right is positive

**03. What does the sensor show?**
Positions increase by 2 m each second; the cart moves right with equal displacements in equal times. Accept observations tied to the displayed values.

**04. Make a prediction**
7 m: another +2 m from 5 m. Assumes the same velocity continues.

**Model / diagram check**




**05. Evidence or a claim?**
E, C, E. Approaching alone does not establish impact.

**06. Test the model**
3 m if the -2 m each second pattern continues. Check measurement quality and whether motion changed.

**07. Ask a question you can investigate**
Examples: Where will it be in a month? Need repeated positions, velocity direction and time. How large is it? Need observations of size. Accept a measurable question and matching evidence.

**08. Choose useful evidence**
The measured times with the track distance allow calculation and repetition. The phrase alone gives no measurable comparison.

**09. Exit check**
12 m, assuming +3 m each second continues. A later position inconsistent with that prediction, beyond measurement variation, challenges the model.

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**10. STRETCH: test the model**
Many answers: continuing gives 14 m; stopping at 2 s gives 8 m. Observe later positions and times.

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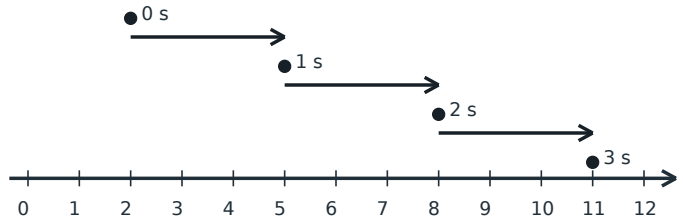
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**Model / diagram check**




Position (m); right is positive

**03. Extract the evidence**
Approach and uncertainty are stated. Impact, a precise arrival date, or a damage estimate is not established.

**04. Fit a simple model**
x = 1 + 3 t in meters, with t in seconds; x(5)=16 m. Assumes constant velocity beyond the observed interval.

**05. Challenge the prediction**
Prediction 16 m lies outside 13.8-14.2 m. Motion may have changed or earlier measurements/model assumptions may be wrong. Do not infer a specific cause without evidence.

**06. Design the observation**
Positions in a stated frame at two known times estimate velocity including direction. Other justified observations accepted; must explain how they improve the prediction.

**07. Compare claims**
Need positions of both bodies at the same time, gravitational/orbital changes, 3 D state and uncertainty. A simplified model alone cannot establish real impact.

**08. Choose useful evidence**
The measured times with the track distance allow calculation and repetition. The phrase alone gives no measurable comparison.

**09. Exit check**
v=-2 m/s; x=12-2(7)=-2 m. Assumes constant velocity. Measure x at a stated later time and compare with the model including uncertainty.

**10. STRETCH: test the model**
One rule x=2+3 t throughout; another follows that rule through 2 s then stays at 8 m. At 4 s: 14 m versus 8 m; future behavior is unmeasured.